

Modelling Feedback Effects with Stochastic Liquidity

Angelika Esser*

Burkart Mönch**

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Abstract

This paper investigates the interactions between stock price movements, the trading strategies of a large trader, and liquidity. Our framework generalizes the model of Frey by introducing a stochastic liquidity factor. We derive a formula for the feedback effect of the large investor's trading strategy and examine the feedback effects on stock prices for positive and contrarian trading strategies. Features of our model are illustrated using Monte Carlo simulation. The main contribution of this paper is the examination of the price process in an economy with stochastic liquidity, such that liquidity shocks can be captured and their consequences can be analyzed. We observe significant liquidity feedback effects on both the price process of the underlying and the trading strategy of the large investor.

Key Words: illiquid markets, large trader models, feedback-effect, stochastic liquidity, asset price process, hedging error

JEL classification: G12.

*School of Business and Economics, Goethe University, Mertonstr. 17, D-60054 Frankfurt am Main, Germany; phone: ++49 (0) 69 798-25101, fax: ++49 (0) 69 798-22788, email: esser@wiwi.uni-frankfurt.de

**School of Business and Economics, Goethe University, Mertonstr. 17, D-60054 Frankfurt am Main, Germany; phone: ++49 (0) 69 798-25109, fax: ++49 (0) 69 798-22788, email: moench@wiwi.uni-frankfurt.de

1 Introduction

The turmoil in financial markets in September 2001 reminds us that a potential lack of liquidity is an important source of risk for portfolio managers. Not only in the small cap but also in the blue chip market liquidity can dry out almost completely, especially when markets collapse. In these situations institutional investors are typically forced to close their positions immediately due to internal or external regulations. If the positions under consideration are substantial in a sense that the trading activity of the institutional investor influences the asset price process considerably, ignoring liquidity issues can result in a dangerous underestimation of the downside risk of a position.

We present a framework in which on the one hand the stock price process is influenced by the trading activity of a large investor and a liquidity factor. On the other hand the dynamics of the stock price and the liquidity factor have an impact on the trading strategy of the large investor. To combine these ideas we model a stochastic liquidity factor which has two main effects: First, it influences the trading strategy of the institutional investor and second, it has an impact on the degree with which the stock price reacts to the trading activity of the large investor.

Furthermore, if we assume that the large trader acts rational, he will recognize the risk associated with the changing degree of liquidity. Thus, he will adjust his trading strategy, and this will in turn have feedback effects on asset prices.

Our approach generalizes both the model of Frey [3] where liquidity is constant and an extension by Patie [5] where liquidity is a deterministic function of the stock price. Modelling liquidity as a stochastic factor first of all enables us to incorporate changes in market depth. Furthermore we can significantly generalize existing models by introducing the concept of liquidity feedback effects. The presence of liquidity feedback effects implies that (i) trading strategies of large investors are affected by the degree of illiquidity and (ii) the sensitivity of stock prices to the trading strategies of large investors can vary due to changes in liquidity.

Why may it be interesting to treat liquidity as an autonomous source of risk? Consider for example an insurance company. If natural disasters occur, the insurance company may have to liquidate substantial amounts of assets in a short time in order to compensate clients for suffered losses. Therefore, a smart risk management in an insurance company will allow the management to invest certain amounts of the portfolio only in highly liquid assets so that the company is able to meet possible obligations in time. Thus, in this example, portfolio managers have to take care of the liquidity risk associated with their investments and have to close positions, if the illiquidity of an asset exceeds a certain threshold.

In our paper, we first analyze the behaviour of stock prices under feedback effects theoretically

and numerically: The effective price process is derived taking feedback effects of trading strategies and liquidity into account. Second, we compare simulation results for the Black-Scholes model [1] (henceforth BS), the constant liquidity model and stochastic liquidity. The results impressively illustrate the importance of a stochastic liquidity factor.

Further, we present implications of our approach for hedging by analyzing the tracking error in our setting. Introducing a stochastic liquidity factor yields an incomplete market. Thus, it is not possible to achieve a tracking error of zero, even when parameters are adjusted due to the effective stock price process as in Frey [3].

The paper is organized as follows: section 2 summarizes the main ideas of Frey's model. The general setting of our framework is presented in section 3. In section 4 we derive the effective dynamics for the underlying asset with stochastic liquidity in order to analyze the feedback effect of the large investor's trading strategy on the stock price dynamics. We demonstrate the effects of stochastic liquidity on stock price processes by simulating asset price movements under different scenarios in section 5. Furthermore, the stock price distributions for BS, constant and stochastic liquidity are compared. Section 6 deals with the tracking error using different hedging strategies. The paper concludes in section 7 with a brief summary and a discussion of issues for further research.

2 The model of Frey

The paper of Frey [3] proposes the following model setup. There exists a risky asset S (the stock) and a risk-free investment earning a zero interest rate (the bond). There are no liquidity effects on the bond, only the underlying S is affected by this source of risk. Further, there is one large investor whose trading strategy influences the price process of the underlying. The underlying follows the stochastic differential equation

$$dS_t = \sigma S_t dW_t^S + \rho S_t d\alpha_t.$$

Here α denotes the number of stocks held by the large trader and $\rho \geq 0$ is a constant liquidity parameter. An increase in ρ means a declining liquidity in the market. For $\rho = 0$ the model represents the standard BS setup with zero drift. $1/(\rho S)$ is called the market depth, i.e. the order size that moves the price by one unit.

The influence of the trading strategy on the price process is discussed in the case of a smooth strategy $\alpha = \phi(S_t, t)$, i.e. $\alpha \in C^{1,1}$. This yields the effective dynamics for the underlying

$$dS_t = b(S_t, t)dt + S_t v(S_t, t) dW_t^S,$$

where

$$\begin{aligned} v(S, t) &= \frac{\sigma}{1 - \rho S \phi_S} \\ b(S, t) &= \frac{\rho S}{1 - \rho S \phi_S} \left(\phi_t + \frac{1}{2} \phi_{SS} S^2 v^2 \right) \end{aligned}$$

assuming $\rho S \phi_S < 1$. Note that volatility has changed from σ to $\frac{\sigma}{1 - \rho S \phi_S}$ due to the strategy ϕ and the liquidity parameter ρ .

As discussed in Frey [3] there are two basic types of trading strategies to consider: On the one hand the large investor can trade using a *positive feedback* strategy, i.e. $\phi_S > 0$. That means he buys the risky asset when the price is increasing, and he sells when the price is declining which reinforces the effect of rising or falling prices, since the true volatility increases in this case. For example in a standard BS model one would use such a strategy to duplicate a convex payoff like a long call. On the other hand the large trader can use a *contrarian feedback* strategy, i.e. $\phi_S < 0$, which means buying stocks when prices drop and vice versa, thus absorbing volatility. This would be the strategy used to duplicate a concave payoff, like a short call.

Apart from the behaviour of stock prices Frey also examines the tracking error, i.e. the deviation of the hedge portfolio from the value of the claim at maturity. He shows that a standard BS delta hedge yields a positive tracking error due to the effective dynamics of the underlying. Taking the adjusted parameters of the effective dynamics of the underlying to derive a partial differential equation (henceforth pde) for the price leads to a tracking error of zero. This will be analysed for the stochastic liquidity framework in section 6.

3 A Stochastic Liquidity Model

In our model the underlying asset price process is assumed to follow the stochastic differential equation

$$dS_t = \sigma S_t dW_t^S + \rho_t S_t d\alpha_t. \quad (1)$$

In contrast to Frey [3] we now assume that ρ_t is stochastic with dynamics given by

$$d\rho_t = \beta(\rho_t, t)dt + \nu(\rho_t, t)dW_t^\rho \quad (2)$$

where the instantaneous covariation is

$$dW_t^S \bullet dW_t^\rho = \gamma dt. \quad (3)$$

A sensible specification for ρ could be a mean-reversion process since we may assume a natural long-run level of liquidity in the market. This liquidity level is approached again after liquidity

shocks. We further assume that the process stays strictly positive for $\rho_0 > 0$. This restricts the choices for the volatility function $\nu(\rho, t)$. One might for example use $\nu(\rho, t) = \nu\rho^{1/2}$.

Rewriting the above dynamics, using a Cholesky-decomposition of the covariance matrix of dS_t and $d\rho_t$ we obtain

$$dS_t = \sigma S_t dW_t + \rho_t S_t d\alpha_t \quad (4)$$

$$d\rho_t = \beta(\rho_t, t)dt + \nu(\rho_t, t)\gamma dW_t + \nu(\rho_t, t)\sqrt{1 - \gamma^2}d\bar{W}_t \quad (5)$$

with a two-dimensional Brownian motion $(W_t, \bar{W}_t)$ (having independent components as usual). Note that α (the number of shares held by the large investor) now not only depends on S and t , but also on the stochastic liquidity factor ρ . So its effects on the trading strategy are twofold: First, α is influenced by changes in S and second, due to our stochastic liquidity setup, α changes with changing liquidity. The impact of S on α corresponds to the positive and the contrarian feedback strategy as the two basic types of trading strategies considered: The first is an increasing function α of S for all ρ , the latter a decreasing function α of S for all ρ . To characterize the impact of ρ on α consider the following scenario: The more illiquid the market the less shares the large trader will hold due to external or internal regulations, no matter whether a positive or contrarian feedback strategy is considered. Thus, a reasonable choice would be a decreasing absolute α -value with respect to ρ (for all S).

What is the gain in modelling achieved by the introduction of a stochastic liquidity factor? Assume that the large trader holds K shares of the asset. Consider the basic strategy that the large trader sells all his holdings of the asset when the price falls below a certain level $\bar{S}$. Furthermore, due to external or internal regulations the large trader may be forced to sell not only in the former case, but also when liquidity decreases. The large trader will close his position if the market becomes highly illiquid, that means when a certain level of illiquidity $\tilde{\rho}$ is reached from below (Remember the example of the insurance company in the introduction.).

We denote the stopping time τ as the first time when a certain price bound $\bar{S}$ is undershot or a certain level of illiquidity $\tilde{\rho}$ is overshot. The downward jump in the asset price process in the first case is by a factor of $1 - \rho_\tau K$. This yields $S_\tau = \bar{S}(1 - \rho_\tau K)$, such that the large trader can sell his K shares only for $KS_\tau = K\bar{S}(1 - \rho_\tau K)$, i.e. he incurs a loss of $K^2\bar{S}\rho_\tau$. In the second case the downward jump in the asset price process is given by $1 - \tilde{\rho}_\tau K$, such that $S_\tau = S_{\tau-}(1 - \tilde{\rho}_\tau K)$, where $S_{\tau-} \geq \bar{S}$. Note that the large trader incurs a loss of $K^2S_{\tau-}\rho_\tau$ in both cases.

As discussed in Schönbucher/Wilmott [6] the loss can be minimized using a sequence of trading times selling only a proportion of the total amount at every point in time. This is of course not very helpful in practice since it is not possible to sell infinitely often in a finite (short) time interval. Besides that, a great number of sales would in practice increase direct transaction costs

like brokerage fees which we neglect in this setting. No matter how clever the strategy chosen, we will always suffer a loss due to illiquidity.

4 Hedging with Feedback Effects

Consider a smooth trading strategy $\alpha := \phi(t, S_t, \rho_t) \in C^{1,2,2}$. An application of Itô's formula leads to the following proposition.

Proposition 1 (Dynamics of the State Variables)

Suppose the trading strategy of the large trader is given by $\phi(t, S_t, \rho_t) \in C^{1,2,2}$. Then the solution to the system of stochastic differential equations (4) and (5) satisfies

$$dS_t = b(t, S_t, \rho_t)dt + v(t, S_t, \rho_t)S_t dW_t + \bar{v}(t, S_t, \rho_t)S_t d\bar{W}_t \quad (6)$$

$$d\rho_t = \beta(\rho_t, t)dt + \nu(\rho_t, t)\gamma dW_t + \nu(\rho_t, t)\sqrt{1 - \gamma^2}d\bar{W}_t, \quad (7)$$

where

$$v(t, S, \rho) = \frac{\sigma}{1 - \rho S \phi_S} + \gamma \frac{\rho \phi_\rho \nu}{1 - \rho S \phi_S} \quad (8)$$

$$\bar{v}(t, S, \rho) = \sqrt{1 - \gamma^2} \frac{\nu \rho \phi_\rho}{1 - \rho S \phi_S} \quad (9)$$

$$b(t, S, \rho) = \frac{\rho S}{1 - \rho S \phi_S} \left[(\phi_t + \beta(\rho_t, t)\phi_\rho + \frac{1}{2}\phi_{\rho\rho}\nu^2) + \frac{1}{2}\phi_{SS}S^2(v^2 + \bar{v}^2) + \nu S \phi_{\rho S} \left(\gamma v + \sqrt{1 - \gamma^2} \bar{v} \right) \right]. \quad (10)$$

The proof is given in Appendix A.

In the case of $\rho \equiv 0$ or $\phi \equiv \text{constant}$ we are in the classical BS scenario with zero drift. If $\rho \neq 0$ and $\phi \neq 0$ the trading strategy of the large trader has an effect on the instantaneous volatilities v and $\bar{v}$, as well as on the total volatility and the correlation between the two processes.

Consider first the special case that liquidity has no impact on trading, i.e. $\phi_\rho \equiv 0$. Then, $\bar{v}$ and the second summand of v will vanish. This means we are close to the scenario of Frey, except for the fact that ρ is stochastic, where feedback effects are only incorporated due to the term $\rho S \phi_S$.

The results are much more complex for $\phi_\rho \neq 0$. Assuming $\rho > 0$ we will now analyze the properties of the derivative ϕ_ρ , to discuss how a change in liquidity influences the trading strategy of the large investor. The more illiquid the market becomes the more the large trader may want to close his position, which means to sell everything he is long or to buy back everything he

is short. In the first case ϕ is monotonically decreasing in ρ , starting with a positive ϕ . In the latter case ϕ is monotonically increasing in ρ starting with a negative ϕ . Thus, ϕ is approaching zero in absolute value as ρ tends to infinity.

Without loss of generality we assume a positive ϕ in the following so that ϕ_ρ should be negative, no matter if a positive feedback or a contrarian trading strategy is considered. From (8) we can see that the sign of $v(t, S, \rho)$ depends on γ . For $\gamma \equiv 0$ the value of v is the same as in the paper of Frey [3]. However, here we have an additional volatility parameter $\bar{v}$ corresponding to the second source of risk $d\bar{W}$ that contributes to total volatility. The parameter $\bar{v}$ is negative since ϕ_ρ is negative.

The instantaneous quadratic variation $d[S]$ is given by (see Appendix A)

$$d[S]_t = S^2 \left(\frac{\sigma^2 + \nu^2 \rho^2 \phi_\rho^2 + 2\gamma\sigma\nu\rho\phi_\rho}{(1 - \rho S \phi_S)^2} \right) dt$$

so that the total instantaneous volatility v_{tot} is equal to

$$v_{tot} = \sqrt{v^2 + \bar{v}^2} = \frac{\sqrt{\sigma^2 + \nu^2 \rho^2 \phi_\rho^2 + 2\gamma\sigma\nu\rho\phi_\rho}}{1 - \rho S \phi_S}. \quad (11)$$

Note that the argument of the square root is non-negative since $|\gamma| \leq 1$.

The instantaneous covariation of the two processes $d[S, \rho]$ is given by

$$d[S, \rho]_t = S\nu \left(\gamma v + \sqrt{1 - \gamma^2 \bar{v}} \right) dt = S\nu \left(\frac{\nu\rho\phi_\rho}{1 - \rho S \phi_S} + \gamma \frac{\sigma}{1 - \rho S \phi_S} \right) dt. \quad (12)$$

Thus, the instantaneous correlation η equals

$$\eta = \frac{d[S, \rho]}{\nu S v_{tot} dt} = \frac{\gamma v + \sqrt{1 - \gamma^2 \bar{v}}}{\sqrt{v^2 + \bar{v}^2}} = \frac{\nu\rho\phi_\rho + \gamma\sigma}{\sqrt{\sigma^2 + \nu^2 \rho^2 \phi_\rho^2 + 2\gamma\sigma\nu\rho\phi_\rho}}. \quad (13)$$

It is important to note in (13) that $\gamma \equiv 0$ does not imply that the increments of the effective stock price process and the liquidity process are uncorrelated. Indeed, in this setup $\eta = 0$ is impossible for a deterministic choice of γ . Even for $\gamma \equiv 0$, there is still some correlation $\eta < 0$ induced by $\rho\phi_\rho$, which is negative since $\phi_\rho < 0$. The correlation η is equal to zero for a stochastic $\gamma = -\nu\rho\phi_\rho/\sigma$, in the special cases of deterministic ρ and $\gamma = 0$, or if a trading strategy independent of ρ and $\gamma = 0$ is considered. In general, η changes randomly over time in our setting due to ρ and $\phi_\rho(t, S, \rho)$. The strongest correlations $\eta = \pm 1$ are not achieved for $\gamma = \pm 1$, but for stochastic $\gamma = \frac{v}{\sqrt{v^2 + \bar{v}^2}}$.

In order to contrast the formal setup of Frey [3] and Patie [5] to our framework (henceforth SLM) we analyze the volatility and correlation structure for different specifications of the respective liquidity related parameters. An overview is given in table 1.

Table 1: Liquidity related parameters in different models

Model	ρ	γ	v	$\bar{v}$	v_{tot}	η
BS	0	0	σ	0	v	0
Frey	const.	0	$\frac{\sigma}{1-\rho S\phi_S}$	0	v	0
Patie	$\rho(S)$	1	$\frac{\sigma+\nu\rho\phi_\rho}{1-\rho S\phi_S}$	0	v	1
SLM	stoch.	0	$\frac{\sigma}{1-\rho S\phi_S}$	$\frac{\nu\rho\phi_\rho}{1-\rho S\phi_S}$	$\sqrt{v^2 + \bar{v}^2} > v$	$\frac{\bar{v}}{\sqrt{v^2 + \bar{v}^2}} < 0$
SLM	stoch.	$\neq 0$	$\frac{\sigma+\gamma\nu\rho\phi_\rho}{1-\rho S\phi_S}$	$\frac{\sqrt{1-\gamma^2}\nu\rho\phi_\rho}{1-\rho S\phi_S}$	$\sqrt{v^2 + \bar{v}^2}$	$\frac{\sqrt{1-\gamma^2}v+\gamma\bar{v}}{\sqrt{v^2 + \bar{v}^2}}$

For $\rho \equiv 0$ or $\phi \equiv \text{constant}$, respectively, we are in the standard BS model without drift, resulting in $\bar{v} \equiv 0$ and $v = \sigma$. The first additional feature is included in Frey's approach where liquidity is represented by a constant ρ (i.e. $d\rho \equiv 0$) implying $\beta \equiv \nu \equiv 0$. This yields $\bar{v} \equiv 0$ and

$$v \equiv v_{tot} = \frac{\sigma}{1 - \rho S\phi_S}$$

resulting in zero correlation η . Especially, this implies that all terms containing ϕ_ρ vanish so that the dependence of the strategy on ρ is of no interest for the effective stock price dynamics.

Second, a deterministic liquidity function $\rho(S)$ instead of constant liquidity is considered, see Patie [5]. Since ρ is a deterministic function of S in this case, the dynamics of ρ are only driven by the first component W of the two-dimensional Brownian motion, implying $\bar{v} \equiv 0$ and $\gamma = 1$. Using Itô the volatility ν of the liquidity process must equal $Sv \frac{d\rho(S)}{dS}$. From this we see that v carries an additional summand of the form $1 \times \frac{\rho\phi_\rho \frac{d\rho(S)}{dS}}{1-\rho S\phi_S}$. The facts that $\bar{v} \equiv 0$ and $\gamma \equiv 1$ yield $\eta \equiv 1$ in this case which depicts the perfect correlation of the increments of the stock price process and the liquidity process. This is inherited from the perfect correlation of the components of the Brownian motion. Thus, the approach of Patie is a special case in our setting with $\gamma = 1$ and a special choice of ν . Note that in this special case the value of η is equal to γ , which is not true in general.

Finally, we discuss the contribution of stochastic liquidity: in this case both, the volatilities v and $\bar{v}$ as well as the instantaneous correlation η between the increments of the stock price process and the liquidity process are influenced by the stochastic liquidity factor for arbitrary γ .

First, consider the case $\gamma = 0$. Then, v is the same as in Frey [3], but in our model we have the additional volatility parameter

$$\bar{v} = \frac{\nu\rho\phi_\rho}{1 - \rho S\phi_S} < 0.$$

The variable $\bar{v}$ incorporates the liquidity feedback effect. It contributes to the total volatility if and only if the trading strategy actually varies with changing liquidity.

Then, the total volatility increases. We obtain a total volatility of

$$v_{tot} = \frac{\sqrt{\sigma^2 + \nu^2\rho^2\phi_\rho^2}}{1 - \rho S\phi_S}.$$

Here, we see a key result of our approach: For $\gamma = 0$ the total volatility of the stochastic liquidity model is greater than the volatility in the deterministic liquidity framework, no matter what type of strategy is used. For $\gamma = 0$ there is a negative correlation between S and ρ , given by

$$\eta = \frac{\nu\rho\phi_\rho}{\sqrt{\sigma^2 + \nu^2\rho^2\phi_\rho^2}} < 0.$$

This term ultimately models liquidity feedback effects. It vanishes for a trading strategy independent of ρ , i.e. for $\phi_\rho \equiv 0$. The negative value of η due to liquidity effects can be interpreted in the following way: If illiquidity is high over a longer period the trader will be forced to close his position which will cause the stock price to drop.

For $\gamma \neq 0$ it is more difficult to analyze the characteristics of the model. The direction of the changes in total volatility (see equation (11)) depends on the sign of γ . The parameter v carries an additional summand describing the liquidity feedback effect, given by ϕ_ρ , and $\bar{v}$ is reduced in absolute values due to $\gamma \neq 0$ (see equations (8) and (9)). The total volatility in (11) is a decreasing function of γ , since $\phi_\rho < 0$. Thus, even for $\gamma \leq 0$ the total volatility is greater than the volatility in Frey's setup. Only for large positive values of γ it is possible that the total volatility is lower than the volatility in Frey's model. This will be the case if

$$1 \geq \gamma > \frac{1}{2\sigma}\nu\rho|\phi_\rho|. \quad (14)$$

For $\gamma \neq 0$ the numerator of the correlation η in equation (13) carries an additional summand $\gamma\sigma$. Thus, η is an increasing function of γ .

Finally, we look at the two basic types of trading strategies with respect to S . A positive feedback strategy, i.e. $\phi_S > 0$, leads to

$$v_{tot} > \sqrt{\sigma^2 + \rho^2\nu^2\phi_\rho^2 + 2\gamma\sigma\nu\rho\phi_\rho}.$$

The expression on the right hand side is greater than or equal to σ if and only if

$$\gamma \leq \frac{1}{2\sigma}\nu\rho|\phi_\rho|.$$

Especially, v_{tot} is greater than σ for $\gamma \leq 0$.

A contrarian feedback strategy (i.e. $\phi_S < 0$) implies that the total volatility satisfies

$$v_{tot} < \sqrt{\sigma^2 + \rho^2 \nu^2 \phi_\rho^2 + 2\gamma\sigma\nu\rho\phi_\rho}$$

which is less or equal to σ if and only if equation (14) is valid.

Let us finally focus on the liquidity feedback effects again: If the trading strategy does not depend on ρ there is no liquidity feedback effect on the asset price process, even in the case of stochastic liquidity. For $\phi_\rho \equiv 0$ we are close to the framework of Frey, except that the liquidity parameter is now stochastic. However, we also have to take into account the instantaneous correlation η between the stock price process and the stochastic liquidity. For $\phi_\rho \equiv 0$ this correlation is equal to the instantaneous correlation γ between the increments of the components of the Brownian motion. Thus, the difference between γ and η is a result of the liquidity feedback effect, exclusively.

5 Numerical Results

5.1 Overview

In this section we present some simulation-based results for three models: a deterministic and a stochastic liquidity model and the BS model (i.e. $\rho \equiv 0$ or $\phi \equiv 0$, respectively) with zero drift as a benchmark. First, BS dynamics are compared to stock price dynamics in a stochastic liquidity setting considering both positive and contrarian feedback strategies. Second, the positive feedback strategy is examined in more detail. Here, we compare the stochastic liquidity setup to the case of a constant positive liquidity parameter.

5.2 Feedback Strategies

The main challenge in using our framework for practical purposes is to find an appropriate functional form for the dependence of the strategy ϕ on the stock price S and the liquidity parameter ρ . In order to simplify the presentation of our example, we restrict ourselves to positive holdings of the asset by the large trader, i.e. $\phi > 0$, at any point in time. The initial holdings ϕ_0 of the asset are assumed to be optimal with respect to ρ and S , in a sense that the large trader has no intention to buy or sell additional units of the asset at time $t = 0$.

Figure 1 shows the dependence of ϕ on ρ and S for the positive feedback strategy.

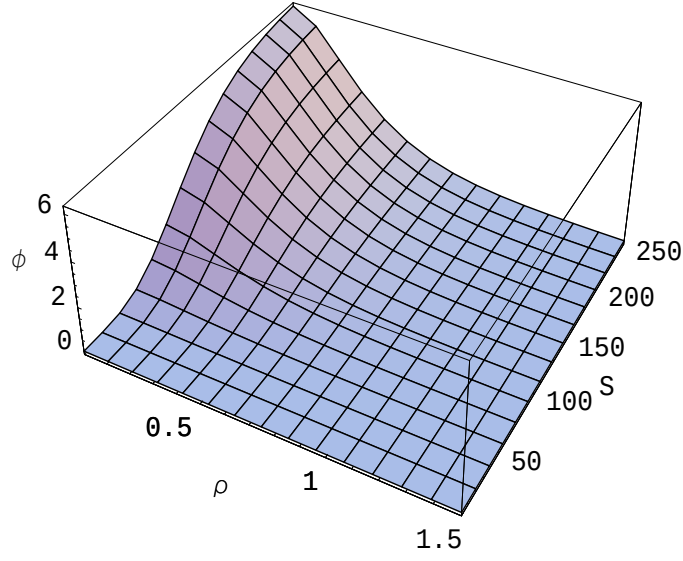


Figure 1: Positive Feedback Strategy

Figure 2 represents the contrarian feedback strategy.

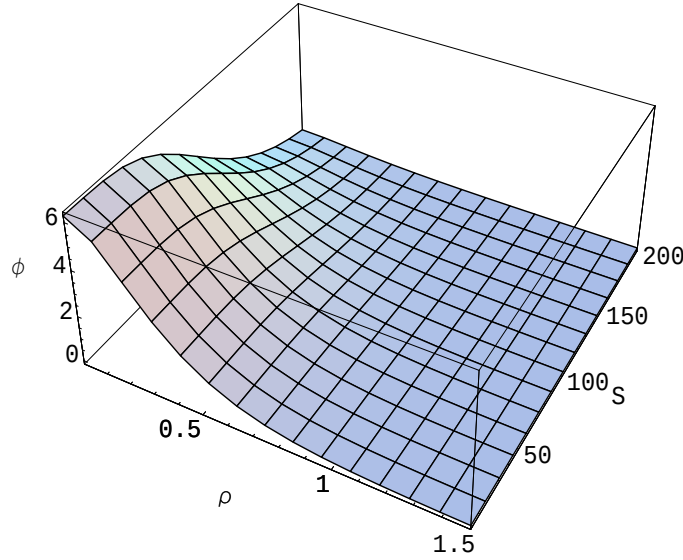


Figure 2: Contrarian Feedback Strategy

To generate the patterns shown in the two figures one might consider the following scenario. If liquidity drops, i.e. ρ increases, the large trader is forced to sell some shares due to external regulations. Consider for example a mutual fund that is allowed to invest only in stocks that

are part of a blue chip index. When the stock becomes illiquid and does therefore not belong to the index any more the fund has to close his position in this asset. Therefore, ϕ is assumed to be monotonically decreasing in ρ for all S and for any feedback strategy, i.e. the derivative of ϕ with respect to ρ is negative. It seems reasonable to assume that for small values of ρ the value of ϕ_ρ is also small in absolute values since changes in small values of ρ have no implications in the market depth of the asset. If ρ increases, i.e. liquidity declines, the trader wants to get rid of his assets, thus ϕ_ρ will increase in absolute values. However, for large values of ρ the absolute value of ϕ_ρ will again decrease and approach zero, since the large trader has already sold almost all of his holdings in the stock.

In order to characterize the relationship between ϕ and S we have to distinguish between the positive feedback and the contrarian feedback strategy.

In the first (second) case, (see figure 1 (2)) the large trader buys (sells) assets as the stock price increases and sells (buys) when the stock price declines. Thus, ϕ is monotonically increasing (decreasing) in S for all ρ in the case of the positive (contrarian) feedback strategy. For very small and very large values of S the changes of the stock holdings of the large trader are negligible when the asset prices vary (similar to the relationship between ρ and ϕ). However, for asset prices in between the absolute value of ϕ_S increases when S increases and a positive (contrarian) feedback strategy is considered.

There is a variety of functional forms for $\phi = \phi(S, \rho)$ that are able to capture the scenario described above. In order to incorporate the behaviour of ϕ as a function of both S and ρ we chose a product separating the strategy with respect to S and ρ :

$$\phi(S, \rho) = a\psi(S)\chi(\rho)$$

A function that is able to model the depicted behaviour is the incomplete gamma function, defined by

$$\Gamma(x, z) = \int_z^\infty e^{-t} t^{x-1} dt.$$

We read $\Gamma(x, z)$ as a function of z for fixed x -values. It is monotonically decreasing to zero for $z \rightarrow \infty$. The 'typical' shape of the incomplete gamma function can for example be seen in figure 2 for fixed S . The incomplete gamma function captures the behaviour of the trading strategy with respect to ρ for fixed S . For the contrarian feedback strategy the same is valid with respect to S for fixed ρ . Since the opposite holds for the positive feedback strategy the functional form has to be adjusted. Therefore we multiply the gamma function by -1 and then shift it to positive quadrant. Thus, we chose the following representation for the dependence on ρ

$$\chi(\rho) = \Gamma(b_1, c_1\rho)$$

and for the dependence on S

$$\psi(S) = \begin{cases} \Gamma(b_2, c_2 S) & \text{(contrarian strategy)} \\ \Gamma(b_2) - \Gamma(b_2, c_2 S) & \text{(positive strategy)} \end{cases}$$

Thus, the contrarian feedback strategy is given by

$$\phi^-(S) = a\Gamma(b_1, c_1 \rho)\Gamma(b_2, c_2 S) \quad (15)$$

whereas the positive feedback strategy is given by

$$\phi^+(S, \rho) = a\Gamma(b_1, c_1 \rho) [\Gamma(b_2) - \Gamma(b_2, c_2 S)]. \quad (16)$$

where $\Gamma(x)$ denotes the standard gamma function.

As an example for the simulation and the plots above we use the following parameter constellation:

$$\begin{array}{llll} b_1 & = & 2.1 & b_2 & = & 6.0 \\ c_1 & = & 5.5 & c_2 & = & 0.05 & a & = & -0.05 \end{array}$$

We assume the initial values $\rho = 0.05$ and $S = 80$. The slope of ϕ with respect to ρ is steepest for ρ around 0.1 and steepest with respect to S for S around 120. Note that in this example the function $\phi(\rho, S)$ is independent of time. This does not cause a problem, since the drift of the underlying processes is zero in the BS setting, and quite small in the setting with a stochastic liquidity parameter.

5.3 Results

5.3.1 General Behaviour of the Paths

To create paths of the underlying and the liquidity parameter Monte Carlo Simulation techniques are used. The stochastic processes are discretized with an Euler scheme with $N = 4000$ steps and $\Delta t = 1/360$. In order to ensure non-negativity and a stationary behaviour of the liquidity process we specify the dynamics for ρ in equation (2) as a square-root-process with a mean-reverting drift component:

$$\beta(\rho, t) = \theta(\kappa - \rho), \quad \nu(\rho, t) = \nu\sqrt{\rho}.$$

In the first picture we compare positive and contrarian strategies with stochastic liquidity to the standard BS case with perfect liquidity and zero drift. That is, for a fixed realization of $(W_t, \bar{W}_t)$ we have plotted the dynamics of equations (4) and (5) for

1. BS: $\rho \equiv 0$, i.e. $\beta \equiv \nu \equiv 0$

2. stochastic liquidity with parameter values

$$\begin{array}{llll} \theta & = & 0.35 & \kappa = \rho_0 = 0.05 & \sigma & = & 0.1 \\ \mu & = & 0.05 & S_0 & = & 80.0 & \gamma & = & 0.0 \\ \nu & = & 0.2 & & & & & & \end{array}$$

i) for $\phi = \phi^-$: contrarian feedback strategy, equation (15)

ii) for $\phi = \phi^+$: positive feedback strategy, equation (16)

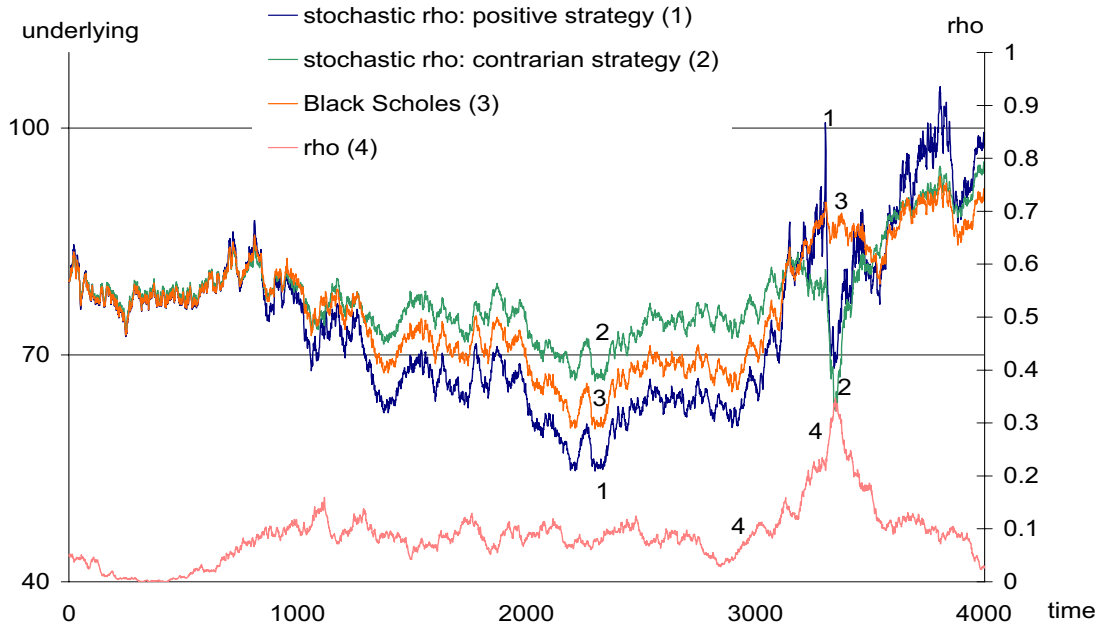


Figure 3: Positive and Contrarian Feedback Strategies versus Black Scholes

Path 3 describes the stochastic process of the underlying S in a BS world with zero drift. Path 4 shows the stochastic liquidity parameter. Path 1 depicts the stock price movement for a positive feedback strategy with a stochastic liquidity parameter. It can be seen that path 1 exceeds path 3 when the BS price is increasing over a longer period. Rising stock prices motivate the large investor to buy additional stocks which will cause the stock price to grow even further. The opposite effect can be observed for a decreasing S , since in this case the large investor wants to get rid of his holdings which will accelerate the decline in the stock price.

The role of the liquidity parameter ρ is more subtle. In fact it can have two different implications: First, all else equal, the trader has to sell stocks if they become more and more illiquid. Second,

if ρ is very high, the stock becomes more volatile, so that a large trader, who follows a positive feedback strategy can cause the stock price to rise to tremendously high values in bullish markets. However, close to maturity of the boom in the market, i.e. when illiquidity exceeds a certain threshold ($\rho \approx 0.3$ in figure 3), the large trader is forced to close its position due to liquidity constraints and the market collapses.

Path 2 describes the stock price if the large investor follows a contrarian feedback strategy. As mentioned above, a positive feedback strategy generates more volatile stock prices compared to BS. In contrast to Frey's result it is in general no longer true in our setup that for a contrarian strategy the volatility of the asset price is reduced compared to BS, as one can see in the right part of figure 3. If the asset becomes very illiquid, the trading activities of the large trader will dominate the stock price dynamics and will have a destabilizing effect compared to BS.

If ρ approaches zero, path 1 and path 2 run parallel to the BS path 3, as one can observe at the beginning of the series.

5.3.2 Positive Feedback Effect

In figure 4 we contrast stock prices simulated in the deterministic liquidity model (path 5) with stock prices generated in the stochastic liquidity model (path 1). As before, path 3 shows the price process of the stock in the BS model without drift. The constant liquidity parameter is set equal to the initial value $\rho_0 = 0.05$ which coincides with the long run mean. The other parameters are chosen as before.

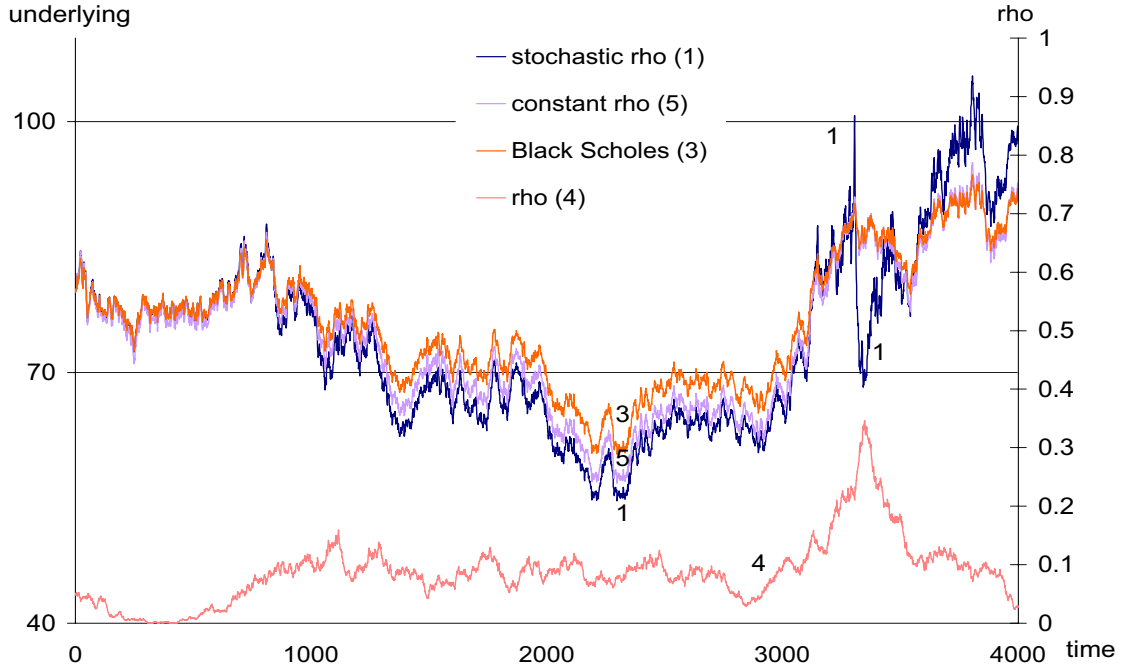


Figure 4: Positive Feedback Strategy

It can be seen that for a constant liquidity parameter the volatility is increased compared to BS (due to the trading activity of the large trader), whereas the general pattern of the stock price movement is similar to the BS path. For this choice of parameters (especially for $\gamma = 0$) the total volatility of the stochastic liquidity model exceeds the one with constant liquidity. As mentioned above, there are additional effects that can reinforce, cancel out, or even turn the direction of the stock price movement compared to the case of deterministic liquidity.

Those features that distinguish our model from other approaches become obvious from time step 3,000 on in figure 4. At this point in time the BS path starts increasing whereas initially path 1 and path 5 are below the BS path 3. After a while they reach the BS path 3 because of the positive feedback strategy. At this point in time, there is a significant increase in the liquidity path, i.e. the market becomes highly illiquid. For high values of ρ the volatility of the asset increases significantly. The asset price process for the stochastic liquidity model is mainly governed by the positive feedback strategy of the large trader. But close to the peak in the liquidity process the large trader starts to sell substantial amounts of the asset. Since illiquidity is still high, this causes a crash in the stock price. Thus, the behaviour of the price process is dominated by the liquidity feedback effect when illiquidity reaches its peak.

This is a typical pattern known from the TMT stocks. Liquidity dries out when a market in a boom is close to its peak. Then an institutional investor starts to liquidate his position and initiates a crash. This phenomenon cannot be modelled in Frey's approach.

One should be aware that one property of the model is the persistence of price or liquidity shocks, if stock price volatility remains low after the shocks. This phenomenon can be observed in the center of the figure 4 for both the stochastic and the deterministic liquidity model. For certain purposes one would rather assume that after a price or a liquidity shock the market recovers immediately and returns to some kind of natural level. In this context one might think of a mean reversion process, that has the BS stock price as a long term mean. To incorporate this, one has to modify the setup fundamentally. This is left for further research.

5.3.3 Stock Price Distributions

Figure 5 shows the stock price distributions for the different models using a positive feedback strategy based on the parameter specification from the previous section. The graphs are generated using an Euler discretization for the simulation and a kernel estimation for the densities.

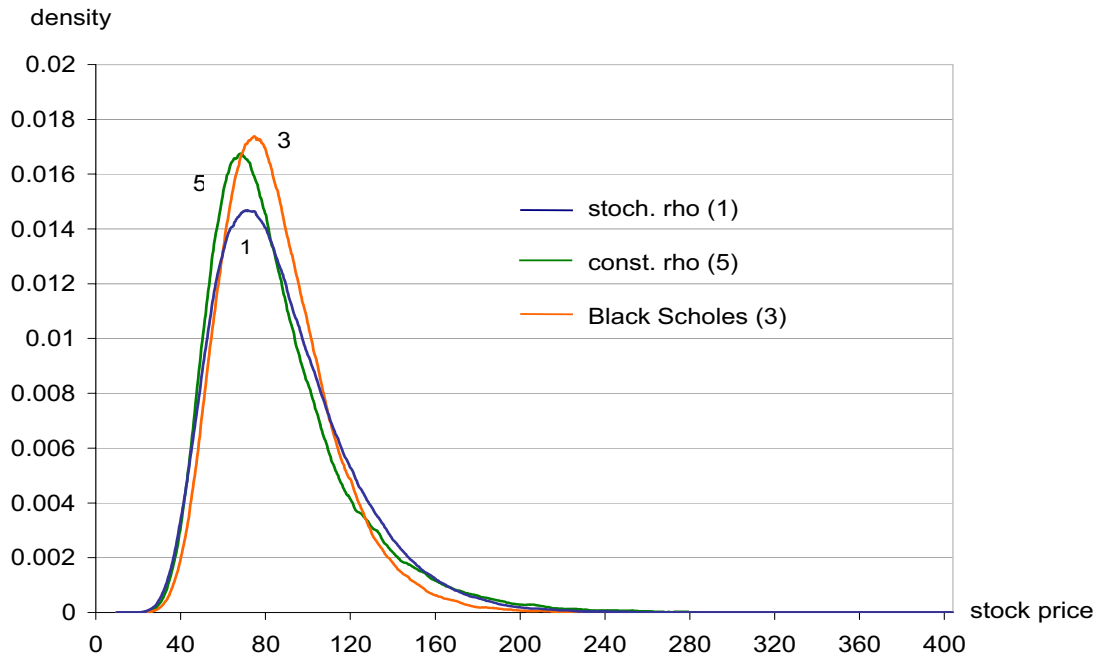


Figure 5: Stock Price Distributions

The BS distribution and the distribution for a constant liquidity parameter have a peak with

almost the same height. The distribution with constant liquidity has slightly more weight in the tails due to higher volatility. The distribution for the stochastic liquidity parameter has a significantly smaller peak and fatter tails compared to the other two due to a higher local volatility. This pattern is similar to stochastic volatility models which is shown for example in Heston [4].

6 Tracking error

The tracking error e_T is defined as the difference between the payoff of the claim $h(S_T)$ and the value V_T of the hedge portfolio at maturity T , i.e.

$$e_T = h(S_T) - V_T.$$

For a perfect hedge the tracking error is identically equal to zero. In the case of a constant $\rho \neq 0$ there is a positive hedging error if the standard BS delta hedge is used. This tracking error is influenced by the trading strategy of the large trader due to the effective price process. It vanishes if $v = \sigma/(1 - \rho S \phi_S)$ is used instead of σ as the volatility parameter (see Frey [3]). Now, we analyze hedging strategies in our setting with a stochastic liquidity parameter ρ .

Assume the large trader follows a BS delta hedge for the claim $h(S_T)$, i.e. the BS value of the claim at time t is given by $u(S, t)$, where u solves the following pde (again assuming a zero interest rate).

$$u_t + \frac{1}{2}\sigma^2 S^2 u_{SS} = 0, \quad u(S, T) = h(S_T) \quad (17)$$

The tracking error is

$$e_T = h(S_T) - V_T = h(S_T) - \left(V_0 + \int_0^T \alpha_t dS_t \right)$$

where $\alpha = u_S$.

In our scenario the following proposition holds:

Proposition 2 (Tracking Error)

The tracking error is given by

$$\begin{aligned} e_T = & \int_0^T u_\rho d\rho + \frac{1}{2} \int_0^T \left(u_{SS} S^2 \left(\sigma^2 \left(\frac{1}{(1 - \rho S u_{SS})^2} - 1 \right) + \frac{\rho^2 \nu^2 u_{S\rho}^2 + 2\gamma \rho \sigma \nu u_{S\rho}}{(1 - \rho S u_{SS})^2} \right) \right) dt \\ & + \frac{1}{2} \int_0^T \nu^2 u_{\rho\rho} dt + \int_0^T S \nu \left(\frac{\nu \rho u_{S\rho} + \gamma \sigma}{1 - \rho S u_{SS}} \right) u_{S\rho} dt. \end{aligned}$$

The proof is given in Appendix B.

In general the tracking error is different from zero, since for non-zero ρ there are feedback effects which are not taken into account using a standard BS delta hedge. For constant ρ we end up with the result of Frey [3], who in particular shows that the tracking error is always positive. For $\rho \equiv 0$, the tracking error vanishes identically.

Here, the hedging error depends on the gamma of the claim u_{SS} and the other partial derivatives u_ρ , $u_{\rho S}$ and $u_{\rho\rho}$. Changes not due to the gamma of the claim are caused by the impact of stochastic liquidity. Thus, even for fixed parameters the sign of the tracking error can be positive or negative depending on the realized paths for S and ρ .

Since the effective price process of the underlying incorporates two sources of risk, it is sensible to correct the hedging strategy for feedback effects. Taking into account both the dynamics of S and ρ with the adjusted parameters from proposition 1 we obtain a tracking error different from the one shown in proposition 2. All but the first term vanish due to the adjusted parameters. The additional risk introduced by the non-traded stochastic liquidity parameter cannot be hedged away. There is no choice of parameters such that the hedging error vanishes identically, since the market is incomplete. The systematic component of ρ , induced by the increment of the ('original') Brownian motion dW^ρ is left, which could only be eliminated using an additional hedging instrument. The following proposition makes this statement more precise.

Proposition 3 (Adjusted Tracking Error)

Assume that the following non-linear pde in two variables admits a solution $u = u(S, \rho, t)$

$$u_t + \frac{1}{2} \left(u_{SS} S^2 \left(\frac{\sigma^2 + \rho^2 \nu^2 u_{S\rho}^2 + 2\gamma\rho\sigma\nu u_{S\rho}}{(1 - \rho S u_{SS})^2} \right) + \nu^2 u_{\rho\rho} \right) + \left(\frac{\nu^2 \rho u_{S\rho} + \nu\gamma\sigma}{1 - \rho S u_{SS}} \right) u_{S\rho} = 0.$$

Then the self-financing trading strategy $\alpha = u_S$ leads to a tracking error e_T for the claim $h(S_T) = u(S_T, \rho, T)$ given by

$$e_T = \int_0^T u_\rho d\rho_t.$$

The sign of the tracking error cannot be determined uniquely.

The proof is given in appendix C.

Note that it could be sensible also to incorporate an u_ρ -term in the pde in order to adjust the mean of e_T . Adding the drift of ρ times u_ρ in the pde would change the tracking error with respect to the drift. In this case we would end up with

$$e_T = \int_0^T u_\rho \nu dW^\rho$$

which is zero in expectation.¹ It is impossible that the tracking error vanishes identically.

There are some problems with this hedging approach. First, it is not obvious, if this non-linear pde admits a sensible solution. Second, the numerical approach will be complicated, since we need a three dimensional grid for solving this pde. Even in the simpler case with constant ρ there are a number of difficulties in numerics to handle, see Patie [5].

Another approach might be to switch over to different hedging strategies. Since the non-traded stochastic liquidity factor is driven by an additional source of risk techniques of incomplete models, which do not provide a perfect hedge, such as superhedging, mean-variance hedging or local risk-minimizing could be useful setting up a hedge portfolio. This may be taken into account for further research.

7 Conclusion

Criteria such as feedback and liquidity effects, hedging error and also computational aspects should be considered in the development of a liquidity model. We have presented a framework that we feel meets these requirements and has proved worthy of investigation.

This paper discusses the impact of the trading strategies of a large investor in a model with stochastic liquidity. It is shown that in the case of a smooth trading strategy the asset price process follows a diffusion process with two sources of risk. The new feature of our model is the liquidity feedback effect. Its impact can be explicitly seen in the form of the volatility parameters of the effective stock price process and the correlation between the asset price process and the liquidity process.

We have analyzed two types of trading strategies: For positive feedback strategies volatility is increased compared to BS for non-positively correlated Brownian motions, but it is in general no longer true that contrarian feedback strategies reduce volatility. When the asset is highly illiquid the trading activity of a large trader, who follows a contrarian feedback strategy, can have a destabilizing effect. We have discussed situations where asset price dynamics are dominated by liquidity shocks, for example consider a rising stock price and increasing illiquidity. Then the trading activity of a large investor can cause a crash due to high illiquidity.

Further, we have examined the stock price distribution of the different models. It turns out that the stochastic liquidity model has fatter tails than the lognormal distribution (BS), a phenomenon known from stochastic volatility models. Finally, we have discussed some issues

¹Note that we assumed here a zero market price of liquidity risk, otherwise we would have to change the measure and adjust the Brownian motion.

related to the computation of tracking errors in this incomplete market model.

Further research could focus on derivative pricing under the effective price processes or the application of hedging strategies from incomplete markets to this setting.

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Appendix

A Proof of Proposition 1

Using Itô for $\phi(t, S_t, \rho_t)$ leads to

$$d\phi = \phi_t dt + \phi_S dS + \phi_\rho d\rho + \phi_{S\rho} d[S, \rho]_t + \frac{1}{2}(\phi_{SS} d[S]_t + \phi_{\rho\rho} \nu^2 dt)$$

where subscripts denote partial derivatives. Plugging this into equation (4) we get

$$\begin{aligned} dS_t = & (\sigma + \rho\phi_\rho\nu\gamma)S_t dW_t + \rho\phi_\rho\nu S_t d\bar{W}_t + \rho_t S_t (\phi_t dt + \phi_S dS + \phi_\rho d\rho + \nu\phi_{S\rho} d[S, \bar{W}]_t + \\ & \frac{1}{2}(\phi_{SS} d[S]_t + \phi_{\rho\rho} \nu^2 dt)) \end{aligned}$$

which implies

$$dS_t = \frac{1}{1 - \rho S \phi_S} \left((\sigma + \rho \phi_\rho \nu \gamma) S_t dW_t + S_t (\rho \phi_\rho \nu \sqrt{1 - \gamma^2}) d\bar{W}_t + \rho S_t \left[(\phi_t + \beta(\rho_t, t) \phi_\rho + \frac{1}{2} \phi_{\rho\rho} \nu^2) dt + \frac{1}{2} \phi_{SS} d[S]_t + \nu \phi_{S\rho} d[S, \bar{W}]_t \right] \right),$$

assuming $\rho_t S_t \phi_S < 1$. Using the trial solution

$$dS_t = b(t, S_t, \rho_t) dt + v(t, S_t, \rho_t) S_t dW_t + \bar{v}(t, S_t, \rho_t) S_t d\bar{W}_t$$

and comparing coefficients leads to

$$\begin{aligned} v(t, S, \rho) &= \frac{\sigma + \rho \phi_\rho \nu \gamma}{1 - \rho S \phi_S} \\ \bar{v}(t, S, \rho) &= \frac{\nu \rho \phi_\rho \sqrt{1 - \gamma^2}}{1 - \rho S \phi_S} \\ b(t, S, \rho) dt &= \frac{\rho S}{1 - \rho S \phi_S} \left[(\phi_t + \beta(\rho_t, t) \phi_\rho + \frac{1}{2} \phi_{\rho\rho} \nu^2) dt + \frac{1}{2} \phi_{SS} d[S]_t + \nu \phi_{S\rho} d[S, \bar{W}]_t \right] \\ &= \frac{\rho S}{1 - \rho S \phi_S} \left[(\phi_t + \beta(\rho_t, t) \phi_\rho + \frac{1}{2} \phi_{\rho\rho} \nu^2) + \frac{1}{2} \phi_{SS} S^2 \frac{\sigma^2 + \rho^2 \nu^2 \phi_\rho^2 + 2\gamma \rho \sigma \nu \phi_\rho}{(1 - \rho S \phi_S)^2} \right. \\ &\quad \left. + \nu \phi_{S\rho} S \frac{\nu \rho \phi_\rho + \gamma \sigma}{1 - \rho S \phi_S} \right] dt \end{aligned}$$

since

$$\begin{aligned} d[S]_t &= S^2 (v^2 + \bar{v}^2) dt \\ &= S^2 \left(\frac{\sigma^2 + \rho^2 \nu^2 \phi_\rho^2 + 2\gamma \rho \sigma \nu \phi_\rho}{(1 - \rho S \phi_S)^2} \right) dt \end{aligned}$$

and

$$\begin{aligned} d[S, \rho]_t &= S \nu (\gamma v + \sqrt{1 - \gamma^2} \bar{v}) dt \\ &= S \nu \left(\frac{\nu \rho \phi_\rho + \gamma \sigma}{1 - \rho S \phi_S} \right) dt \end{aligned}$$

B Proof of proposition 2

The large trader is assumed to use the standard BS delta hedge. Then, the tracking error is given by

$$e_T = h(S_T) - \left(u(0, S_0) + \int_0^T u_S dS_t \right). \quad (18)$$

Since the delta hedge is a smooth strategy, the underlying asset is driven by a two-dimensional Itô diffusion, as shown in proposition 1. Using Itô for u leads to

$$\begin{aligned} h(S_T) &= u(T, S_T) = u(0, S_0) + \int_0^T u_S dS_t + \int_0^T u_\rho d\rho_t + \int_0^T u_t - \frac{1}{2} u_{SS} \sigma^2 S^2 dt \\ &\quad + \int_0^T \left[\frac{1}{2} \left(u_{SS} S^2 (v^2 + \bar{v}^2) + \nu^2 u_{\rho\rho} \right) + (\nu S v \gamma + \nu S \bar{v} \sqrt{1 - \gamma^2}) u_{S\rho} \right] dt. \end{aligned} \quad (19)$$

Substituting this into (18) and using the standard BS-pde (17) to eliminate the partial derivative of u with respect to t leads to

$$\begin{aligned} e_T &= \int_0^T u_\rho d\rho_t + \int_0^T \left[\frac{1}{2} \left(u_{SS} S^2 (v^2 + \bar{v}^2 - \sigma^2) + \nu^2 u_{\rho\rho} \right) + S \nu \left(v \gamma + \bar{v} \sqrt{1 - \gamma^2} \right) \right] dt \\ &= \int_0^T u_\rho d\rho + \frac{1}{2} \int_0^T \left[u_{SS} S^2 \left(\sigma^2 \left(\frac{1}{(1 - \rho S u_{SS})^2} - 1 \right) + \frac{\rho^2 \nu^2 u_{S\rho}^2 + 2 \gamma \rho \sigma \nu u_{S\rho}}{(1 - \rho S u_{SS})^2} \right) \right] dt \\ &\quad + \frac{1}{2} \int_0^T \nu^2 u_{\rho\rho} dt + \int_0^T S \nu \left(\frac{\nu \rho u_{S\rho} + \gamma \sigma}{1 - \rho S u_{SS}} \right) u_{S\rho} dt. \end{aligned} \quad (20)$$

C Proof of Proposition 3

The proof follows the lines of the previous proposition. The result of equation (19) together with the non-linear pde in proposition 3 shows that everything except the first term in equation (20) cancels out, eliminating u_t as before. Thus, the tracking error using a delta-hedge is given by

$$e_T = \int_0^T u_\rho d\rho_t.$$